

Minimax Estimation

Outline

- 1) Minimax risk, estimator
- 2) Least favorable priors
- 3) Examples

Minimax risk

Last idea for choosing an estimator: worst-case risk

$$\underset{\delta}{\text{minimize}} \quad \sup_{\theta} R(\theta; \delta)$$

The minimum achievable sup-risk is called the minimax risk of the estimation problem

$$r^* = \inf_{\delta} \sup_{\theta} R(\theta; \delta)$$

An estimator δ^* is called minimax if it achieves the minimax risk, i.e.

$$\sup_{\theta} R(\theta; \delta^*) = r^*$$

Game theory interpretation: (Minimax game)

- 1) Analyst chooses estimator δ
- 2) Nature chooses parameter θ to max. risk

NB: Nature chooses θ adversarially, not X

Compare to Bayes ("Maximin" game)

- 1) Nature chooses prior $\triangleleft$ (mixed strategy)
- 2) Analyst chooses estimator to min. (avg) risk

We will look for Nature's Nash-equil. strategy in this problem

Least Favorable Priors

Key observation: average-case risk $\leq$ worst-case risk

For proper prior Λ , the Bayes risk is

$$r_{\Lambda} = \inf_{\delta} \int R(\theta; \delta) d\Lambda(\theta) \\ \leq \inf_{\delta} \sup_{\theta} R(\theta; \delta) = r^*$$

If δ_{Λ} Bayes then $r_{\Lambda} = \int R(\theta; \delta_{\Lambda}) d\Lambda(\theta)$

$\Rightarrow$ Bayes risk of any Bayes estimator
lower bounds r^*

Least favorable prior Λ^* gives best
lower bound: $r_{\Lambda^*} = \sup_{\Lambda} r_{\Lambda}$

Sup-risk of any estimator upper bounds r^*

$$r_{\Lambda} \leq r_{\Lambda^*} \leq r^* \leq \sup_{\theta} R(\theta; \delta) \\ \begin{matrix} \uparrow & & \uparrow \\ \text{(any } \Lambda) & & \text{(any } \delta) \end{matrix}$$

Idea: try to match upper & lower bounds

Theorem Let Λ be a prior with Bayes estimator δ_Λ

If $r_\Lambda = \sup_{\theta} R(\theta; \delta_\Lambda)$, then δ_Λ is minimax and Λ is LF, with $r^* = r_\Lambda$.

If δ_Λ unique Bayes (up to a.s.), also unique minimax.

Proof Any other δ :

$$\sup_{\theta} R(\theta; \delta) \geq \int R(\theta; \delta) d\Lambda(\theta)$$

$$\begin{aligned} & \geq \int R(\theta; \delta_\Lambda) d\Lambda(\theta) \\ & \text{strict if} \\ & \text{unique Bayes} \end{aligned}$$

$$= r_\Lambda$$

$$= \sup_{\theta} R(\theta; \delta_\Lambda) \quad \text{by assumption}$$

$$\text{Hence } r_\Lambda \leq r^* \leq \sup_{\theta} R(\theta; \delta_\Lambda) = r_\Lambda \quad \square$$

Gives checkable conditions:

1) Bayes est. with constant $R(\theta; \delta_\Lambda) \Rightarrow$ minimax

2) If $R(\theta; \delta_\Lambda)$ attains max everywhere on $\text{supp}(\Lambda)$

NB: Showing $r_\Lambda = \mathbb{E}_\Lambda R(\theta; \delta)$ const is meaningless since we have integrated out θ !

Ex $X \sim N(0, 1)$ $|\theta| \geq 1$ ($\Theta = (-\infty, -1] \cup [1, \infty)$)

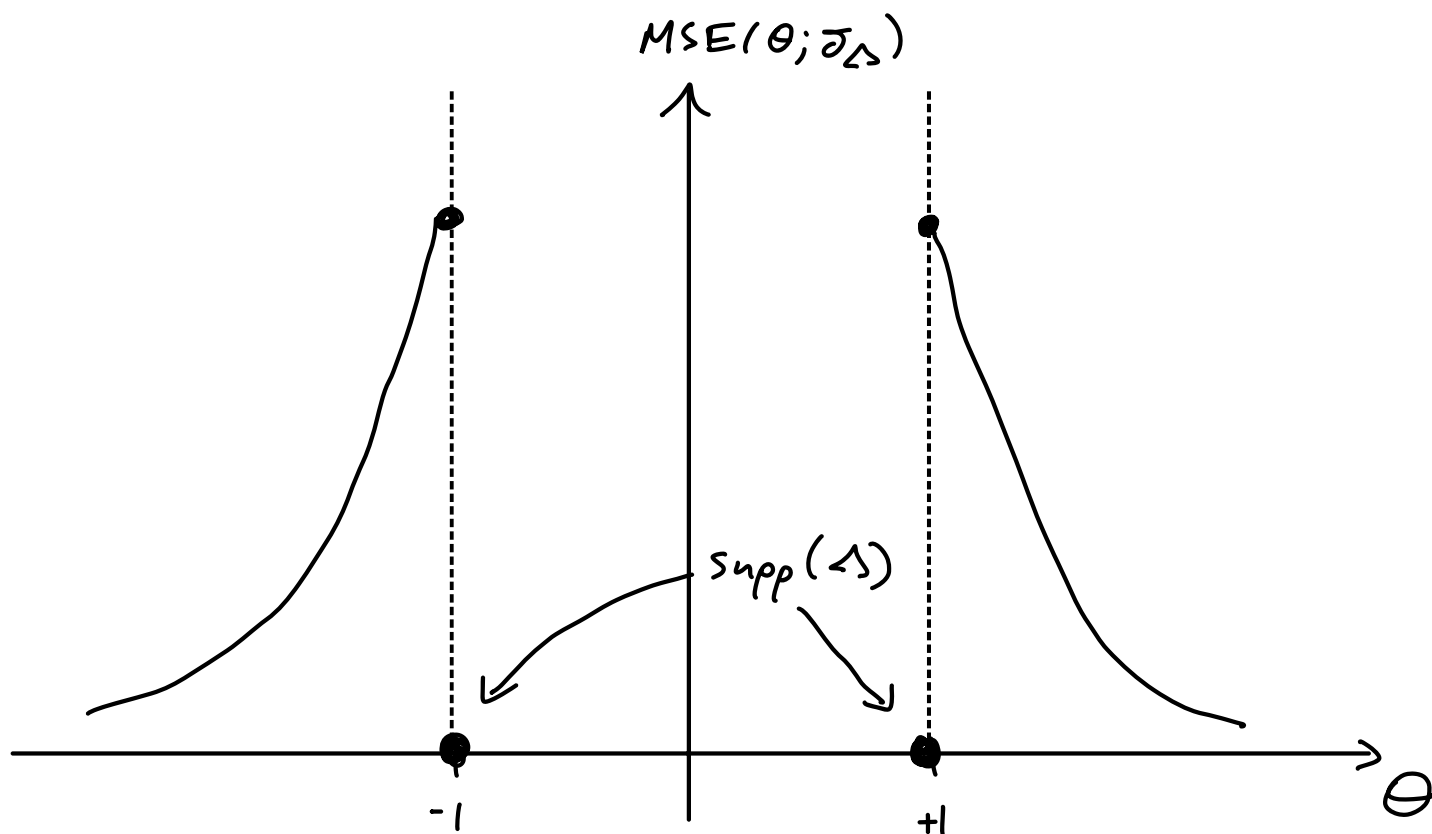
Estimate $g(\theta) = \text{sgn}(\theta) = \begin{cases} +1 & \theta > 0 \\ -1 & \theta < 0 \end{cases}$

Try $\Delta = \text{Unif}(\{-1, +1\})$ ($\Delta(\{-1\}) = \Delta(\{+1\}) = 1/2$)

$$\mathbb{P}_{\Delta}(\theta = 1 \mid X=x) = \frac{\phi(x-1)}{\phi(x-1) + \phi(x+1)}$$

$$\mathbb{E}_{\Delta}(g(\theta) \mid X=x) = \frac{\phi(x-1) - \phi(x+1)}{\phi(x-1) + \phi(x+1)}$$

$$= \frac{e^x - e^{-x}}{e^x + e^{-x}} = \tanh(x)$$



Example (Binomial)

$X \sim \text{Binom}(n, \theta)$, estimate θ , sq. err.

Try $\text{Beta}(\alpha, \beta)$, hope to get one with const. risk

Try $\alpha = \beta$: $\delta_\alpha(X) = \frac{X + \alpha}{n + 2\alpha}$ (symmetric)

$$\mathbb{E}_\theta \delta_\alpha(X) = \frac{n\theta + \alpha}{n + 2\alpha} \Rightarrow \text{Bias}(\theta; \delta_\alpha) = \frac{\alpha(1 - 2\theta)}{n + 2\alpha}$$

$$\text{Var}_\theta \delta_\alpha(X) = \frac{\text{Var}_\theta(X)}{(n + 2\alpha)^2} = \frac{n\theta(1 - \theta)}{n + 2\alpha}$$

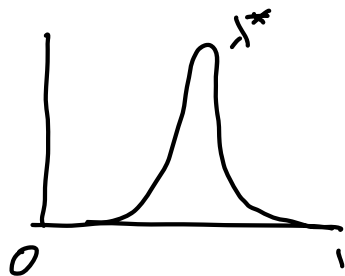
$$\begin{aligned} \text{MSE}(\theta; \delta_\alpha) &= (2\alpha + n)^{-2} [\alpha^2(1 - 2\theta)^2 + n\theta(1 - \theta)] \\ &= (2\alpha + n)^{-2} \left[\alpha^2 + \underbrace{(n - 4\alpha^2)}_{\alpha^* = \sqrt{n}/2} \theta(1 - \theta) \right] \end{aligned}$$

$$\text{MSE}(\theta; \delta_{\alpha^*}) \equiv \frac{n/4}{(n + \sqrt{n})^2} = r^*$$

$$\Rightarrow \delta_{\alpha^*}(X) = \frac{\sqrt{n}/2 + X}{\sqrt{n} + n} \quad \text{minimax}$$

Q: why concentrated at $\theta = 1/2$?

$$\Delta^* = \text{Beta}(\sqrt{n}/2, \sqrt{n}/2) \quad \text{LF:}$$



Least Favorable Sequence

Sometimes there is no least favorable prior

Def: A sequence $\Lambda_1, \Lambda_2, \dots$ is LF
if $r_{\Lambda_n} \rightarrow \sup_{\Lambda} r_{\Lambda}$

Thm: Suppose $\Lambda_1, \Lambda_2, \dots$ is a prior sequence
and δ satisfies $\sup_{\theta} R(\theta; \delta) = \lim_n r_{\Lambda_n}$

Then δ is minimax and $\Lambda_1, \Lambda_2, \dots$ is LF

Proof Other est. $\tilde{\delta}$, $\forall n$,

$$\begin{aligned} \sup_{\theta} R(\theta; \tilde{\delta}) &\geq \int R(\theta; \tilde{\delta}) d\Lambda_n(\theta) \\ &\geq r_{\Lambda_n} \end{aligned}$$

$$\begin{aligned} \Rightarrow \sup_{\theta} R(\theta; \tilde{\delta}) &\geq \sup_n r_{\Lambda_n} \\ &\geq \lim_n r_{\Lambda_n} \\ &= \sup_{\theta} R(\theta; \delta) \end{aligned}$$

Hence $\lim_n r_{\Lambda_n} \leq r^* \leq \sup_{\theta} R(\theta; \delta) = \lim_n r_{\Lambda_n} \quad \square$

Ex $X \sim N_d(\theta, I_d)$. Est. θ , use MSE

$$\delta_0(x) = x \quad \text{unbiased, } MSE = d$$

Prior $\Delta_n = N_d(0, nI_d) \rightsquigarrow \delta_{\xi_n}(x) = (1 - \xi_n)x$ $\xi_n = \frac{1}{n+1}$

$$MSE(\theta; \delta_{\xi_n}) = \xi_n^2 \|\theta\|^2 + (1 - \xi_n)^2 d$$

$$r_{\Delta_n} = \mathbb{E} MSE(\theta)$$

$$= \xi_n^2 nd + (1 - \xi_n)^2 d$$

$$= d \left(\frac{n}{(1+n)^2} + \frac{n^2}{(1+n)^2} \right)$$

$$\rightarrow d$$

So $\Delta_1, \Delta_2, \dots$ is LF,

$\delta_0(x)$ is minimax (but inadmissible?)

$$r^* = d$$

Bounding minimax risk

Our theorem gives an idea of how to bound r^* for a problem:

Upper bound: If δ is any estimator then

$$r^* \leq \sup_{\theta} R(\theta; \delta) \quad (= \text{if } \delta \text{ minimax})$$

Lower bound: If Δ is any prior then

$$r^* \geq \int R(\theta; \delta_{\Delta}) d\Delta(\theta) \quad (= \text{if } \Delta \text{ LF})$$

Minimax estimators are very hard to find but minimax bounds are often used in stat theory to characterize hardness (esp. lower)

Ex: Propose practical estimator δ , find Δ for which r_{Δ} close to $\sup_{\theta} R(\theta; \delta)$ (or same rate, or cugs asymptotically)

$\Rightarrow$ Conclude δ can't be improved "much" (*)

Ex: Quantify hardness of a problem by its minimax rate in some asy. regime.

Caveat: A problem might be easy throughout most of par. space but very hard in some bizarre corner you never encounter in practice!